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Submitted

Your grade: 100%

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Your latest: 100%

Your highest: 100%

24 Aug 2024 10:22 AM

At least 10 min. After you submit

Graded Assignment: Working with 2D Medical Images

Instructions

COOCH

Instructions

Ready to review what you've learned before starting the assignment? I'm here to help.

To complete this quiz, open MATLAB and import the file "test_data.mat" included in the course files' Data\Xray Images folder. Then answer the questions below.

Assignment details

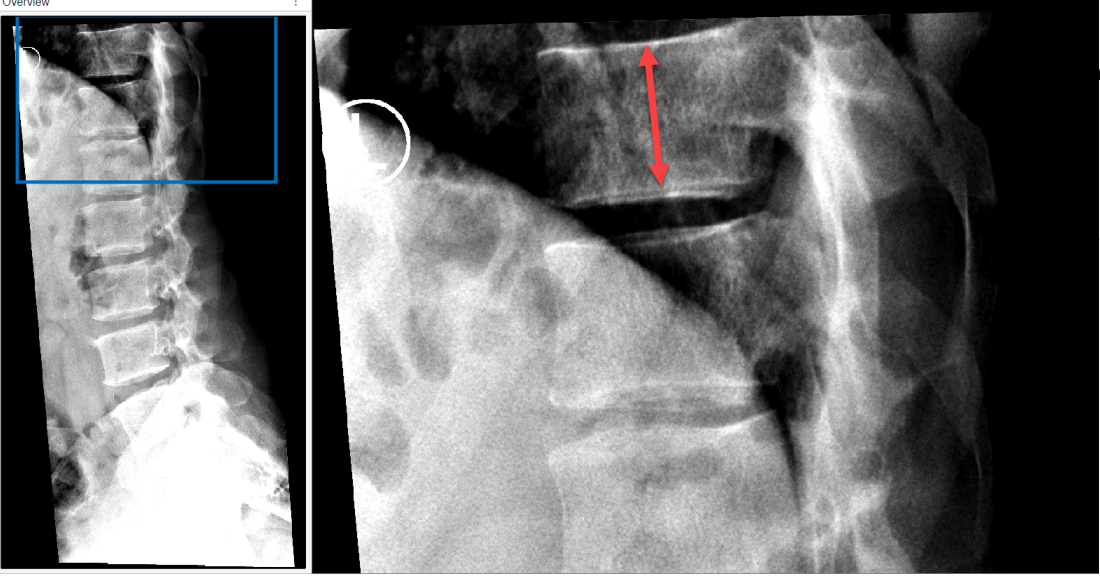
Due Aug 5, 11:59 PM IST

Attempts Unlimited

1 / 1 point

1. Using the Image Viewer app, right-click on the vertebrae at the indicated location. What is the approximate height in mm?

Submitted



32.1184

Correct

The pixel spacing property is used to convert the measured distance in pixels to millimeters.

2. What is the maximum intensity value stored in the Pixel property?

4095

Correct

Pixel intensity values range from 0 to 4095.

3. The pixel intensity values in the back x-ray do not use the full range of values supported by the data type (uint16). Which function will scale the intensity values to use the full range?

1 / 1 point

☐ linspace

☐ resample

☐ ind2double

☒ imad3uint

☐ Correct

This function preserves the image datatype while stretching the range of intensities to use the full range of values supported by the datatype.

4. Import the image using the medicalImage function. Which property contains the real-world size of a single pixel?

1 / 1 point

☒ PixelSpacing

☐ Pixels

☐ SpatialUnits

☐ Correct

This property contains a 2-element vector of the form [Δx Δy].

5. What is the height of this x-ray image in millimeters?

1 / 1 point

☐ 3026

☐ 0.1390

☒ 420.9

☐ Correct

Real-world height is found by multiplying the number of rows in the PixelA property array by the Δy value in the PixelSpacing property.